

4 In this question you must show detailed reasoning.

The roots of the cubic equation $x^3 - 3x^2 + 19x - 17 = 0$ are α , β and γ .

(a) Find a cubic equation with integer coefficients whose roots are $\frac{1}{2}(\alpha - 1)$, $\frac{1}{2}(\beta - 1)$ and $\frac{1}{2}(\gamma - 1)$. **[4]**

(b) Hence or otherwise solve the equation $x^3 - 3x^2 + 19x - 17 = 0$. **[3]**